# MILITARY SPECIFICATION

## BOX ASSEMBLY, PNEUMATIC LOCK CONTROL

This specification is approved for use by the U.S. Army Armament, Munition and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This military specification covers one type of control box assembly used in the operation of the pneumatic lock incorporated in the Weapon Access Delay System program.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-8140

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PRODUCT DRAWINGS

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| | |
|---|---|
| 9352620 | Plate, Valve |
| 9352672 | Manifold "B" Subassembly |
| 9354395 | Comparator/Tubing Assembly |
| 9378139 | Logic Enclosure |
| 9378143 | Receptacle |
| 9378150 | Pneumatic Lock Control Box Assembly, |

INSPECTION EQUIPMENT DRAWINGS

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

| | | |
|---|---|---|
| 9354302 | - | Key Blank |
| 9378144 | - | Gasket |
| 9378146 | - | Encoder/Tubing Assembly |

(Copies of drawings, and publications required by manufacturers in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the contracting officer.)

2.2. Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulation unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components. Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on the component drawing. The contractor shall have available verifiable proof (i.e., objective evidence (see 6.4)) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

3.2 Assembly. The control box assembly shall comply with all requirements specified on drawings (dwg.) 9378150, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.



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3.3 First article. When specified, a sample control box assembly shall be subjected to first article inspection (see 4.3 and 6.2).

3.4 Functional tests.

3.4.1 Code comparator assembly. The code comparator assembly (dwg. 9354271) shall be pneumatically tested to determine that the output pressure, at each of the nine output ports, is not less than 2 pounds per square inch gage (psig) with a pressure of 65 ± 5 psig applied, in turn, to input ports A and B. During this test, no obvious audible nitrogen gas leaks shall be noted between the cover plate and the manifold subassembly. Testing shall be as specified in 4.5.1.1.

3.4.2 Control box assembly. The control box assembly shall be capable of meeting the requirements of Table I with an input pressure of 120 ± 5 psig. During this test, the control box assembly shall not fail to function correctly, and there shall be no evidence of deterioration or damage that may impair its intended operation. Testing shall be as specified in 4.5.1.2.

TABLE I. Functional test requirements for the control box assembly.

| Test point | Pressure (psig) | Time (sec) | Test condition |
|---|---|---|---|
| a. Output hose assembly (PORT 1) | 80 ± 2 | 6-12 | LOCK button pressed |
| b. Output hose assembly (PORT 1) | 60 min | 1.5-3.0 | LOCK button held in; READ button pressed; LOCK button released |
| c. Output hose assembly (PORT 2) | 80 ± 2 | 1.5-3.0 | LOCK button pressed |
| d. Output hose assembly (PORTS 3,4,5,6) | 80 ± 2 at peak | 6-12 | Blank key in encoder; READ button pressed. Blank key removed; READ button pressed |
| e. Valve V5 indicator | -- | 40-60 | Blank key in encoder; READ button pressed |
| f. Volume Chamber C1 | 0 ± .5 | -- | Blank key in encoder; READ button pressed |



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3.5 High temperature. The control box assembly shall withstand a high temperature of 105° ± 5°F, for not less than 6 hours with no evidence of deterioration or damage that may impair its intended operation. While at the high temperature (at the end of the high-temperature exposure period), the control box assembly shall provide an output pressure of not less than 70 psig, and there shall not be any obvious malfunctions, during lock and unlock functions. Testing shall be as specified in 4.5.2.

3.6 Low temperature. The control box assembly shall withstand a low temperature of 0° ± 5°F, for not less than 6 hours with no evidence of deterioration or damage that may impair its intended operation. While at the low temperature (at the end of low-temperature exposure period), the control box assembly shall provide an output pressure of not less than 70 psig, and there shall not be any obvious malfunctions, during lock and unlock functions. Testing shall be as specified in 4.5.3.

3.7 Serialization of the valve plate. A serial number shall be assigned by the contractor to each valve plate produced (see dwg. 9352620). In no event, shall serial numbers be duplicated, nor shall they give any indication of classified production quantities.

3.8 Workmanship. All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. All pneumatic hoses must be free of kinks.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.


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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. Where the contract requires first article inspection, the contractor shall submit two consecutively produced control box assemblies to the testing facility designated in the invitation for bids or request for proposal.

4.3.2 Inspection to be performed. The first article inspection samples shall be inspected for the defects specified in 4.4.2.1, using inspection methods contained therein, and for all the requirements of Section 3, as applicable. The control box assembly test requirements of Section 3 are listed in Table II, and the tests shall be performed in the listed sequence. The functional tests of the code comparator assembly, (see 3.4.1) shall be performed before assembling into the control box assembly. Each item shall be tested with the test methods specified in 4.5. Inspection shall be performed with the acceptance inspection equipment specified in 4.4.4. Should any item fail to comply with any of the applicable requirements, the first article samples shall be rejected. The Government reserves the right to terminate its inspections upon any failure of the first article samples to comply with the stated requirements. This series of tests, listed in Table II, is not considered destructive.



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TABLE II. First article tests.

| Test | Requirement paragraph |
|---|---|
| a. Functional tests<br>1. Control box assembly | 3.4.2 |
| b. High temperature | 3.5 |
| c. Low temperature | 3.6 |

4.4 Quality conformance inspection.

4.4.1 Inspection lot. A lot or batch and its formation, size and presentation is described in MIL-A-48078. Accordingly, a lot shall mean an inspection lot, and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.4.2 Product inspection examination. A sample shall be randomly selected from each inspection lot in accordance with the major defect or the minor defect table as applicable (see Table III). One defect found in the sample, selected for either the major or the minor classification, shall reject the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspections are performed. The term, "SMTE," where used, shall mean any standard measuring and test equipment suitable for the test being performed. The term "TMC," where used, shall mean tool manufacture control (see 6.5).

TABLE III. Inspection lot sampling.

| Major defect table | | Minor defect table | |
|---|---|---|---|
| Lot size | Sample size | Lot size | Sample size |
| 1-5 | all | 1-5 | all |
| 6-10 | 5 | 6-10 | 4 |
| 11-15 | 10 | 11-15 | 8 |
| 16-25 | 15 | 16-25 | 11 |
| 26-35 | 22 | 26-35 | 14 |
| 36-50 | 26 | 36-50 | 16 |
| 51-70 | 30 | 51-70 | 18 |
| 71-100 | 33 | 71-100 | 19 |
| 101 and over | 37 | 101 and over | 20 |

4.4.2.1 Classification of defects. Critical, major, and minor defects are defined in MIL-A-48078.



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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF CHARACTERISTICS

MIL-B-63956 (AR)

| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.1.1 | Plate, Valve | | 9352620 |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical: | None defined. | | | |
| Major: | | | | |
| 101. | Location of corner mounting holes (#1, #2, #269, and #286) | Table III | 3.2 | Gage |
| 102. | Diameter of corner mounting holes (#1, #2, #269, and #286) | Table III | 3.2 | Gage |
| 103. | Location of holes in plate (excluding 4 corner mounting holes) | Table III | 3.2 | Gage |
| Minor: | | | | |
| 201. | Thickness, min. | Table III | 3.2 | Gage |
| 202. | Protective finish inadequate or improper | Table III | 3.2 | Visual |
| 203. | Marking missing, improper, or illegible | Table III | 3.2 | Visual |
| 204. | Serialization missing, improper,or illegible | Table III | 3.7 | Visual |
| 205. | Evidence of poor workmanship | Table III | 3.8 | Visual |

NOTES:



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# QUALITY CONFORMANCE INSPECTION
## CLASSIFICATION OF CHARACTERISTICS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.1.2 | Manifold "B" Sub-assembly | 1 OF 1 | 9352672 |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical: | None defined. | | | |
| Major: | | | | |
| 101. | Component missing, damaged, incorrect or improperly assembled | Table III | 3.2 | Visual |
| 102. | Location of holes (18) incorrect | Table III | 3.2 | SMTE |
| 103. | Diameter of holes (18) incorrect | Table III | 3.2 | SMTE |
| 104. | Location of holes (2) incorrect | Table III | 3.2 | SMTE |
| 105. | Diameter of holes (2) incorrect | Table III | 3.2 | SMTE |
| 106. | Depth of holes (18) incorrect | Table III | 3.2 | SMTE |
| 107. | Coaxiality of holes (18) not met | Table III | 3.2 | SMTE |
| 108. | Alignment of manifolds incorrect | Table III | 3.2 | SMTE |
| 109. | Relative movement in plane between manifolds greater than specified | Table III | 3.2 | SMTE |
| 110. | Protective finish on excepted areas | Table III | 3.2 | Visual |
| 111. | Key Groove slot incorrect (2 places) (7.6 + .01) and (0.135 + .010) | Table III | 3.2 | SMTE |
| Minor: | | | | |
| 201. | Protective finish missing, incomplete or incorrect | Table III | 3.2 | Visual |
| 202. | Marking missing, illegible, incorrect, in improper place or improper depth | Table III | 3.2 | Visual |
| 203. | Evidence of poor workmanship | Table III | 3.8 | Visual |

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# QUALITY CONFORMANCE INSPECTION
## CLASSIFICATION OF CHARACTERISTICS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.1.3 | Comparator/Tubing Assembly | 1 OF 1 | 9354395 |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical: | None defined. | | | |
| Major: | | | | |
| 101. | Component missing, damaged, incorrect or improperly assembled | Table III | 3.2 | Visual |
| 102. | Hose length incorrect (9 places) | Table III | 3.2 | SMTE |
| 103. | Function test requirements improper | Table III | 3.4.1 | 4.5.1.1 |
| Minor: | | | | |
| 201. | Evidence of poor workmanship | Table III | 3.8 | Visual |

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

MIL-B-63956 (AR)

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.1.4 | Enclosure, Logic | 1 OF 1 | 9378139 |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical: | None defined. | | | |
| Major: | | | | |
| 101. | Location of holes (6) incorrect | Table III | 3.2 | SMTE |
| 102. | Location of holes (4) incorrect | Table III | 3.2 | SMTE |
| 103. | Diameter of holes (4) incorrect | Table III | 3.2 | SMTE |
| 104. | Location of hole incorrect | Table III | 3.2 | SMTE |
| 105. | Diameter of hole incorrect | Table III | 3.2 | SMTE' |
| 106. | Location of tapped hole incorrect | Table III | 3.2 | SMTE |
| 107. | Pitch diameter (Max) of tapped hole incorrect | Table III | 3.2 | SMTE |
| 108. | Minor diameter (Min) of tapped hole incorrect | Table III | 3.2 | SMTE |
| 109. | Protective finish on excepted areas | Table III | 3.2 | Visual |

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.



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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF CHARACTERISTICS

MIL-B-63956 (AR)

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.1.5 | Receptacle | 1 OF 1 | 9378143 |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical: | None defined. | | | |
| Major: | | | | |
| 101. | Location of tapped holes(25) incorrect | Table III | 3.2 | SMTE |
| 102. | Pitch diameter (Max) of holes (25) incorrect | Table III | 3.2 | SMTE |
| 103. | Minor diameter (Min) of holes (25) incorrect | Table III | 3.2 | SMTE |
| 104. | Thickness incorrect | Table III | 3.2 | SMTE |
| 105. | Flatness not within requirement (2 surfaces) | Table III | 3.2 | SMTE |
| Minor: | | | | |
| 201. | Surface finish incorrect | Table III | 3.2 | Visual |
| 202. | Chamfer missing (all edges) | Table III | 3.2 | Visual |
| 203. | Chamfer on all holes (25) and both surfaces in excess of requirement | Table III | 3.2 | SMTE |
| 204. | Evidence of poor workmanship | Table III | 3.8 | Visual |

NOTES:



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# QUALITY CONFORMANCE INSPECTION
## CLASSIFICATION OF CHARACTERISTICS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.1.6 | Control Box Assembly | 1 OF 1 | 9378150 |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical: | None defined. | | | |
| Major: | | | | |
| 101. | Key retainer missing, improper or improperly assembled | Table III | 3.2 | Visual |
| 102. | Any part missing, damaged, improper, or improperly assembled (including logic enclosure, pneumatic logic assembly, pigtail assembly, output hose assembly (6), bullet hub (2), and breather (2)) | Table III | 3.2 | Visual |
| Minor: | | | | |
| 201. | Evidence of poor workmanship | Table III | 3.8 | Visual |

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4.4.3 Quality conformance inspection tests. The contractor shall perform all of the tests specified in Table IV. The tests may be performed in any sequence.

TABLE IV. Quality conformance tests.

| Test | Test classification |
|---|---|
| a. Functional tests: | |
| 1. Code comparator assembly | Major |
| 2. Control box assembly | Major |
| b. Low temperature | Major |

4.4.3.1 Functional test: code comparator assembly (see 3.4.1) control box assembly (see 3.4.2), and low temperature (see 3.6). Each code comparator assembly, and control box assembly shall be subjected to these tests, as applicable. Should any code comparator assembly, or control box assembly fail to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

4.4.4 Acceptance inspection equipment.

4.4.4.1 Major defects and tests. Inspection shall be performed with the acceptance inspection equipment, operating instructions, and calibration procedures designed or specified by the contractor. The contractor shall obtain approval of such equipment designs, operating instructions, and calibration procedures prior to use on the contract (see 6.4).

4.4.4.2 Minor defects. The acceptance inspection equipment operating instructions, and calibration procedures used by the contractor for minor classification inspection shall be approved by the Government representative responsible for acceptance inspection.

4.5 Methods of inspection. The tolerances specified in this specification are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to equipment used. Unless otherwise specified, the tests shall be conducted at room temperature (77° ± 10°F).

4.5.1 Functional tests. The pneumatic source to be used for all functional tests shall be a supply of dry, filtered, compressed nitrogen gas.

4.5.1.1 Code comparator assembly. The code comparator assembly shall be functionally tested as follows:



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(A) A regulated nitrogen gas supply shall be adjusted to a pressure of 65±5 psig. This pressure shall be applied to the A-slot input port (located opposite slot A, near the center of the assembly). The output pressure at each of the nine output ports (5 on one side and 4 on the other side of the assembly), shall be a minimum of 2 psig.

(B) Repeat step (A), except apply the input pressure to the B-slot input port (located opposite slot B, near center of the assembly).

The code comparator assembly shall comply with the requirements of 3.4.1 during this test.

4.5.1.2 Control box assembly. The following test conditions apply to the functional test of the control box assembly:

(A) Each control box assembly shall be functionally operated a few times before any pressure steady state (void initial chart over ride dip) measurements are taken.

(B) Unless otherwise specified, the control box assembly shall be functioned with no key in the encoder and with a blank key (dwg. 9354302) in the B slot of the code comparator.

(C) Unless otherwise specified, all pressure measurements shall be recorded with a strip chart recorder coupled with a pressure transducer. The pressure transducer used shall have an equivalent internal volume of one cubic inch. The strip chart recorder used shall have a minimum chart advanced speed of 0.133 inch/sec or 8 ft/min. High temperature and low temperature measurements shall be made by means of a suitable pressure gage. The pressure recorder(s) or gages shall be attached to the output hose assemblies at the top of the control box assembly.

(D) The connection between the control box assembly and the pressure transducer shall be 1/8" ID tubing, 5' ± 2" long.

(E) Where a momentary press of the LOCK or of the READ button is specified, it shall be for a duration of not greater than 1/2 second.



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(F) Whenever the READ button is pressed, nitrogen gas will flow through the encoder for approximately 2 seconds. Do no proceed to the next step until this flow has stopped.

(G) Encoder/tubing assembly (9378146) shall be properly fastened to the receptacle of the pigtail assembly on the side of the control box assembly. Gasket (9378144) must be used at this connection. Torque diametrically opposed captive screws to 20-25 in/LBS. Secure the encoder to a test fixture/stand with "READ" and "LOCK" facing the viewer which permits the keys to be dropped in place. Plug the "L" fitting on the encoder/tubing assembly.

(H) The control box assembly shall be secured to a test fixture/stand in the verticle position.

4.5.1.2.1 Function Test. Perform the functional test in the following sequences:

(A) A regulated nitrogen gas supply shall be adjusted to a supply pressure of 120 PSIG and connected to the T fitting on the attached encoder/tubing assembly.

(B) Momentarily press the LOCK button and check thestatic pressure, and its duration (at 60 psig), at output hose assembly (PORT 1).

(C) Press and hold the LOCK button; then momentarily press the READ button. When the nitrogen gas flow stops, release the LOCK button. The pressure at output hose assembly (PORT 1) will drop from an initial value (approximately 80 psig) to a lower value and then back to the initial value. Measure this lower value. The read- cycle time shall be from the moment the pressure reaches 70 $\pm$ psig to where it returns to 70 $\pm$ psig.

(D) Momentarily press the LOCK button. Check the pressure rise time at output hose assembly (PORT 2) by measuring time from the 0 psig point to the 78 psig point.

(E) Insert a blank key into the encoder, and momentarily press the READ button. Within a 20-second period after the nitrogen gas flow has stopped, remove the blank key and momentarily press the READ button again. Check the pressure at (unlock) output hose assembly (PORTS 3, 4, 5, & 6). Check the application time (at 60 psig) at (unlock) output hose assembly (PORTS 3, 4, 5, & 6).

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(F) Before starting this test, wait about 60 seconds, after the end of test (E), to ensure that the indicator on valve V5 has returned to its original position. Insert a blank key into the encoder and momentarily press the READ button. Check enable functioning by timing the motion of the indicator at the top center of valve V5 as it withdraws until it resets.

(G) Remove restrictor R1 from volume chamber C1. Insert a blank key into the encoder and momentarily press the READ button. Check for pressure of the C1 volume chamber, with a pressure gage.

The control box assembly shall comply with the requirements of 3.4.2 during this test.

4.5.2 High temperature. Test conditions (A) through (C) and (E) of the first paragraph of 4.5.1.2 apply to the high-temperature test of the control box assembly. Perform the high-temperature test in the following sequence:

(A) Place the control box assembly, and its regulated nitrogen gas supply into a test chamber. A regulated nitrogen gas supply shall be adjusted to a supply pressure of 120 PSIG.

(B) Raise the temperature of the chamber to 105° ± 5°F. The control box assembly shall remain at this temperature for not less than 6 hours.

(C) After the 6-hour exposure, connect the regulated nitrogen gas supply to the T fitting of the attached encoder/tubing assembly.

(D) Momentarily press the LOCK button and check the pressure at output hose assembly (PORT 1)

(E) Insert a blank key into the encoder, and momentarily press the READ button. Within a 20-second period after the nitrogen gas flow has stopped, remove the blank key and momentarily press the READ button again. check the pressure at output hose assembly (PORTS 3, 4, 5, & 6)

The control box assembly shall comply with the requirements of 3.5 during this test.



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4.5.3 Low temperature. Test conditions (A) through (C) and (E) through (H) of the first paragraph of 4.5.1.2 apply to the low temperature test of the control box assembly. Perform the low temperature test in the following sequence:

(A) Place the control box assembly, and its regulated nitrogen gas supply, into a test chamber. A regulated nitrogen gas supply shall be adjusted to a supply pressure of 120 PSIG.

(B) Lower the temperature of the chamber to 0° ± 5°F. The control box assembly shall remain at this temperature for a minimum period of 6 hours.

(C) After the 6-hour exposure, connect the regulated nitrogen gas supply to the T fitting o the attached encoder/tubing assembly.

(D) Momentarily press the LOCK button and check the pressure at output hose assembly (PORT 1).

(E) Insert a blank key into the encoder, and momentarily press the READ button. Within a 20-second period after the nitrogen gas flow has stopped, remove the blank key and momentarily press the READ button again. check the pressure at output hose assembly (PORTS 3, 4, 5, & 6).

(F) After test completion, each control box assembly shall be heated to 105° ± 5°F for a period of not less than 6 hours to remove all internal moisture.

The control box assembly shall comply with the requirements of 3.6 during this test.

## 5. PACKAGING

5.1 Preservation, packaging, packing, and marking. The control box assembly shall be preserved, packaged, packed, and marked in accordance with P9378150 or as specified in the contract or purchase order.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The box assemblies covered by this specification are intended for use in pneumatic lock components for the weapon access delay system.



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6.2 Acquisition requirement. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of control box assembly being procured, part number and nomenclature.

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. When first article is required (see 3.3).

e. Applicable methods of packaging and packing (see 5.1).

6.3 Objective evidence.

6.3.1 General. Contractors inspection records and documents verifying compliance with applicable dwg. and specification requirements.

6.3.2 Materials. A statement (certification), supported by test data, that all materials produced or purchased by the contractor meets all requirements when such material is controlled by contractural documents.

6.4 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration procedures from Commander, U.S. Army Armament, Research, Development and Engineering Center, ATTN: AMSMC-QAN-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5 Tool manufacture control (TMC). When TMC is specified as a method of inspection, it shall mean that the manufacturing tool setup (e.g., stamping die, automatic tape control machine, etc.) may be used to accept the parts being manufactured. Tow consecutive parts, which comply with drawing dimensions, shall constitute proper initial tool control acceptance. In addition, two consecutive parts shall be examined, as specified in the applicable paragraph in the classification of defects, whenever the manufacturing tool requires maintenance.

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6.6 Subject term (key word) listing.

code comparator
control box assemly
encoder
coded key
tool manufacture control
acceptance inspection

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 8140-A805)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-B-63956 (AR) | 23 February 1990 |

3. DOCUMENT TITLE

BOX ASSEMBLY, PNEUMATIC LOCK CONTROL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION |
|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) |
| | e. DATE SUBMITTED (YYMMDD) |

7. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code) |
|---|---|
| US ARMY ARDEC<br>STANDARDIZATION OFFICE | (1) Commercial (201) 724-6671<br>(2) AUTOVON 880-6671 |
| c. ADDRESS (Include Zip Code) | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: |
| ATTN: SMCAR-BAC-S<br>PICATINNY ARSENAL, NJ 07806-5000 | Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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